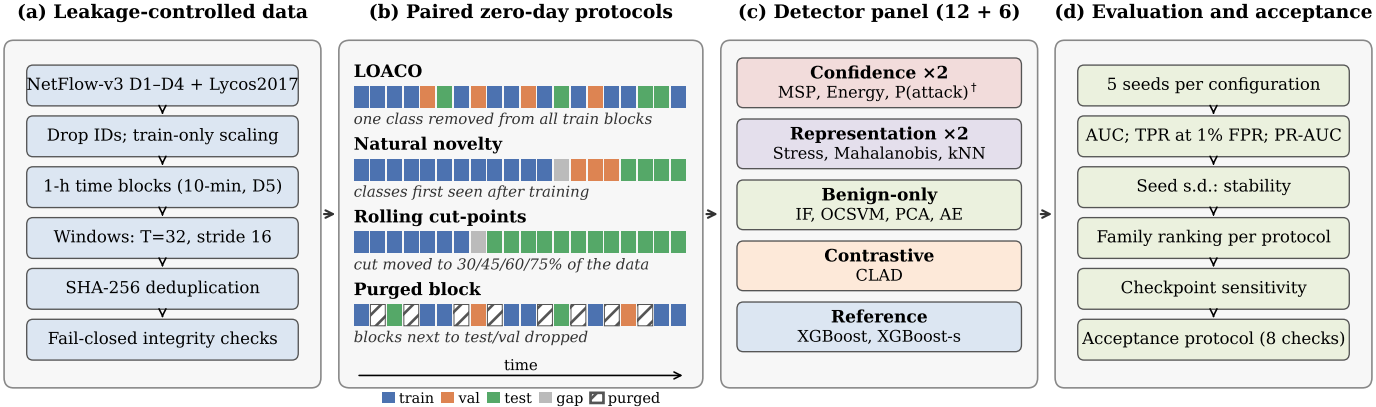

(a) Leakage-controlled data
NetFlow-v3 D1–D4 + Lycos2017
Drop IDs; train-only scaling
1-h time blocks (10-min, D5)
Windows: T=32, stride 16
SHA-256 deduplication
Fail-closed integrity checks
(b) Paired zero-day protocols
LOACO
one class removed from all train blocks
Natural novelty
classes first seen after training
Rolling cut-points
cut moved to 30/45/60/75% of the data
Purged block
blocks next to test/val dropped
time
train
val
test
gap
purged
(c) Detector panel (12 + 6)
Confidence ×2
MSP, Energy, P(attack)†
Representation ×2
Stress, Mahalanobis, kNN
Benign-only
IF, OCSVM, PCA, AE
Contrastive
CLAD
Reference
XGBoost, XGBoost-s
(d) Evaluation and acceptance
5 seeds per configuration
AUC; TPR at 1% FPR; PR-AUC
Seed s.d.: stability
Family ranking per protocol
Checkpoint sensitivity
Acceptance protocol (8 checks)